

EVERY EXPERT WAS ONCE A BEGINNER



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LINUX COMMANDS YOU MUST KNOW AS A DEVOPS ENGINEER



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Linux Commands Cheat Sheet for DevOps Engineers

This cheat sheet provides a comprehensive reference of essential Linux commands frequently used by DevOps engineers for system administration, automation, and troubleshooting.

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1. System Information & Monitoring

Command	Description	Example
<code>uname -a</code>	Show system/kernel info.	<code>uname -a</code>
<code>uptime</code>	System uptime and load average.	<code>uptime</code>
<code>top</code>	Real-time process and system usage (CPU/memory).	<code>top</code>
<code>htop</code>	Interactive process and system usage (CPU/memory).	<code>htop</code>
<code>free -h</code>	Memory usage in human-readable form.	<code>free -h</code>
<code>df -h</code>	Disk usage of mounted filesystems.	<code>df -h</code>
<code>du -sh *</code>	Disk usage of files/directories (current path).	<code>du -sh /var/log</code>
<code>vmstat 1</code>	CPU, memory, and process statistics.	<code>vmstat 1</code>
<code>iostat -xz 1</code>	Detailed CPU & disk I/O (from sysstat).	<code>iostat -xz 1</code>
<code>lscpu</code>	Display CPU architecture information.	<code>lscpu</code>
<code>cat /proc/cpuinfo</code>	Display CPU information.	<code>cat /proc/cpuinfo</code>
<code>lsblk</code>	List block devices.	<code>lsblk</code>
<code>fdisk -l</code>	Manipulate disk partition table.	<code>sudo fdisk -l</code>
<code>date</code>	Print or set the system date and time.	<code>date</code>

2. Process & Service Management

Command	Description	Example
<code>ps aux \ grep <process></code>	Search running process.	<code>ps aux \ grep nginx</code>
<code>kill -9 <pid></code>	Kill a process forcefully.	<code>kill -9 12345</code>
<code>systemctl status <service></code>	Check service status.	<code>systemctl status apache2</code>
<code>systemctl restart <service></code>	Restart service.	<code>systemctl restart nginx</code>
<code>journalctl -u <service></code>	Logs of a specific service.	<code>journalctl -u sshd.service</code>
<code>top</code>	Monitor system processes in real-time.	<code>top</code>
<code>htop</code>	Interactive process viewer (often needs installation).	<code>htop</code>
<code>kill</code>	Terminate processes by PID.	<code>kill 12345</code>
<code>killall</code>	Terminate processes by name.	<code>killall nginx</code>
<code>bg</code>	Send a process to the background.	<code>command & bg</code>
<code>fg</code>	Bring a background process to the foreground.	<code>fg</code>
<code>nohup</code>	Run a command immune to hang-ups.	<code>nohup ./script.sh &</code>

3. User & Permission Management

Command	Description	Example
<code>whoami</code>	Current user.	<code>whoami</code>
<code>id</code>	User and group info.	<code>id</code>
<code>groups</code>	Show group memberships.	<code>groups</code>
<code>chmod 755 file</code>	Change file permissions.	<code>chmod 755 script.sh</code>
<code>chown user:group file</code>	Change ownership.	<code>chown www-data:www-data /var/www/html</code>
<code>sudo -i</code>	Become root shell.	<code>sudo -i</code>
<code>chgrp</code>	Change group ownership.	<code>chgrp dev_team file.txt</code>
<code>umask</code>	Set default permissions for new files/directories.	<code>umask 022</code>

4. File & Directory Management

Command	Description	Example
<code>ls -lh</code>	List files with sizes.	<code>ls -lh /home/user</code>
<code>ls -lrt</code>	Sort by modification time.	<code>ls -lrt /var/log</code>
<code>find /path -name "*.log"</code>	Find files by name.	<code>find /var/log -name "*.log"</code>
<code>grep "ERROR" /var/log/syslog</code>	Search inside files.	<code>grep "ERROR" /var/log/syslog</code>
<code>tail -f /var/log/syslog</code>	Follow log updates live.	<code>tail -f /var/log/nginx/access.log</code>
<code>less /var/log/messages</code>	Scroll logs easily.	<code>less /var/log/messages</code>
<code>pwd</code>	Print the current working directory.	<code>pwd</code>
<code>ls</code>	List files and directories.	<code>ls -la</code> (list all in long format)
<code>cd</code>	Change directory.	<code>cd /var/log</code>
<code>mkdir</code>	Create a new directory.	<code>mkdir my_project</code>
<code>rm</code>	Remove files or directories.	<code>rm -rf old_dir/</code> (force remove recursively)
<code>cp</code>	Copy files and directories.	<code>cp file.txt /backup/file.txt</code>
<code>mv</code>	Move or rename files and directories.	<code>mv old_name.txt new_name.txt</code>
<code>touch</code>	Create an empty file or update timestamp.	<code>touch new_file.txt</code>

5. Networking & Connectivity

Command	Description	Example
<code>ping 8.8.8.8</code>	Test connectivity.	<code>ping 8.8.8.8</code>
<code>curl -I https://example.com</code>	Check HTTP headers/status.	<code>curl -I https://google.com</code>
<code>wget https://example.com/file</code>	Download file.	<code>wget https://example.com/archive.zip</code>
<code>netstat -tulnp</code>	Listening ports and processes (legacy).	<code>netstat -tulnp</code>
<code>ss -tulnp</code>	Listening ports and processes (modern).	<code>ss -tulnp</code>
<code>telnet <host> <port></code>	Check if port is open.	<code>telnet localhost 80</code>
<code>dig example.com</code>	DNS resolution.	<code>dig google.com</code>
<code>nslookup example.com</code>	DNS resolution (alternative).	<code>nslookup google.com</code>
<code>traceroute example.com</code>	Trace route to host.	<code>traceroute google.com</code>
<code>ip</code>	Show/manipulate routing, devices, policy routing.	<code>ip addr show</code>
<code>ifconfig</code>	Display/configure network interfaces (legacy).	<code>ifconfig eth0</code>
<code>nc</code>	Netcat, a networking utility for reading from and writing to network connections.	<code>nc -vz localhost 80</code>
<code>iptables</code>	Administer IP packet filter rules.	<code>sudo iptables -L</code>

Command	Description	Example
<code>firewalld</code>	Dynamic firewall management tool.	<code>sudo firewall-cmd --list-all</code>

6. Package Management

Debian/Ubuntu (APT)

Command	Description	Example
<code>apt update && apt upgrade -y</code>	Update package lists and upgrade installed packages.	<code>sudo apt update && sudo apt upgrade -y</code>
<code>apt install <package></code>	Install new packages.	<code>sudo apt install nginx</code>
<code>apt remove <package></code>	Remove packages.	<code>sudo apt remove apache2</code>
<code>apt search <package></code>	Search for packages.	<code>apt search docker</code>
<code>dpkg</code>	Debian package management utility.	<code>dpkg -i package.deb</code>

RHEL/CentOS (YUM/DNF)

Command	Description	Example
<code>yum install <package></code>	Install packages (legacy).	<code>sudo yum install httpd</code>
<code>dnf update -y</code>	Update packages (modern).	<code>sudo dnf update -y</code>
<code>yum remove <package></code>	Remove packages (legacy).	<code>sudo yum remove mariadb</code>
<code>yum search <package></code>	Search for packages (legacy).	<code>yum search git</code>
<code>rpm</code>	RPM package management utility.	<code>rpm -ivh package.rpm</code>

7. Archive & Compression

Command	Description	Example
<code>tar -czvf backup.tar.gz /path/to/dir</code>	Create compressed archive.	<code>tar -czvf my_backup.tar.gz /var/www/html</code>
<code>tar -xzvf backup.tar.gz</code>	Extract archive.	<code>tar -xzvf my_backup.tar.gz</code>
<code>zip -r backup.zip /path</code>	Create zip archive.	<code>zip -r my_archive.zip /home/user/docs</code>
<code>unzip backup.zip</code>	Extract zip archive.	<code>unzip my_archive.zip</code>

8. SSH & File Transfer

Command	Description	Example
<code>ssh user@host</code>	Connect to remote server.	<code>ssh ubuntu@my-server.com</code>
<code>scp file user@host:/path</code>	Copy file via SSH.	<code>scp local.txt user@remote:/tmp/</code>
<code>rsync -avz /src/ user@host:/dst/</code>	Sync files efficiently.	<code>rsync -avz /var/www/html/ user@remote:/var/www/html/</code>

9. Disk & Storage

Command	Description	Example
<code>lsblk</code>	List block devices/disks.	<code>lsblk</code>
<code>mount /dev/sdb1 /mnt</code>	Mount a device.	<code>sudo mount /dev/sdb1 /mnt/data</code>
<code>umount /mnt</code>	Unmount a device.	<code>sudo umount /mnt/data</code>
<code>fdisk -l</code>	Partition information.	<code>sudo fdisk -l</code>

10. Containers (Docker/K8s essentials)

Command	Description	Example
<code>docker ps -a</code>	List all containers.	<code>docker ps -a</code>
<code>docker logs -f <container></code>	Follow container logs.	<code>docker logs -f my-app-container</code>
<code>docker exec -it <container> bash</code>	Shell inside container.	<code>docker exec -it my-db-container bash</code>
<code>kubectl get pods -A</code>	Get all pods across namespaces.	<code>kubectl get pods -A</code>
<code>kubectl describe pod <pod> -n <namespace></code>	Show pod details.	<code>kubectl describe pod my-pod-xyz -n default</code>
<code>kubectl logs -f <pod></code>	Follow pod logs.	<code>kubectl logs -f my-pod-xyz</code>

11. Version Control

Command	Description	Example
<code>git clone</code>	Clone a repository into a new directory.	<code>git clone <repo_url></code>
<code>git pull</code>	Fetch from and integrate with another repository or a local branch.	<code>git pull origin main</code>
<code>git commit</code>	Record changes to the repository.	<code>git commit -m "Initial commit"</code>
<code>git push</code>	Update remote refs along with associated objects.	<code>git push origin main</code>
<code>git status</code>	Show the working tree status.	<code>git status</code>
<code>svn checkout</code>	Checkout a working copy from a repository.	<code>svn checkout <repo_url></code>
<code>svn update</code>	Bring changes from the repository into the working copy.	<code>svn update</code>
<code>svn commit</code>	Send changes from the working copy to the repository.	<code>svn commit -m "Updated files"</code>

12. Automation & Scheduling

Command	Description	Example
<code>crontab</code>	Schedule recurring tasks.	<code>crontab -e</code> (edit cron jobs)
<code>ansible</code>	Simple IT automation system.	<code>ansible all -m ping</code>
<code>ansible-playbook</code>	Run Ansible playbooks.	<code>ansible-playbook site.yml</code>

13. Text Processing

Command	Description	Example
cat	Concatenate files and print on the standard output.	cat file.txt
head	Output the first part of files.	head -n 10 file.txt
tail	Output the last part of files.	tail -f /var/log/syslog (follow log in real-time)
grep	Print lines matching a pattern.	grep "error" /var/log/syslog
sed	Stream editor for filtering and transforming text.	sed 's/old/new/g' file.txt
awk	Pattern scanning and processing language.	awk '{print \$1}' file.txt
cut	Remove sections from each line of files.	cut -d':' -f1 /etc/passwd
more	Page through text one screenful at a time.	more large_file.txt
less	Page through text, allowing backward movement.	less large_file.txt